Name: ______________________ Date: ____________

Add and Subtract Polynomials

In 1911, an Italian handyman hid in a broom closet overnight and walked out of the Louvre with this legendary portrait hidden under his smock.

Match your answer in the solution bank to find the letter for each blank.

Follow the directions for each set:

Simplify:

(688) $(-5x^3 + 4x^2) + (3x^3 - 2x^2 - x)$

(812) $(3x^2 - 4x) + (2x^2 + 5x)$

(509) $(a^2 - 3a + 2) + (5a^2 + a - 8)$

(904) $(7z - 9) + (2z^2 - 3z + 4)$

Translate and simplify:

(147) Find the sum of $(2m^2 - m + 4)$ and $(m^2 + 6)$

(234) Find the sum of $(4y^3 + 2y)$ and $(y^3 - 7y + 1)$

(351) Find the sum of $(p^4 - 2p^2)$ and $(3p^4 + p^2 + p)$

Solution Bank

Solution	*Letter*
$5x^2 + x$	[M]
$6a^2 - 2a - 6$	[N]
$2z^2 + 4z - 5$	[S]
$5y^3 - 5y + 1$	[O]
$6a^2 - 4a - 6$	[Z]
$-2x^3 + 2x^2 - x$	[L]
$4p^4 - p^2 + p$	[I]
$5y^3 + 9y + 1$	[W]
$3m^2 - m + 10$	[A]
$5x^2 - 9x$	[X]

___ ___ ___ ___   ___ ___ ___ ___

812 234 509 147   688 351 904 147

Name: ________________________ Date: ____________

Algebra 1: Intermediate

Add and Subtract Polynomials

In 1990, two thieves dressed as police officers tied up guards and stole $500 million worth of art from this Boston institution, executing the largest property theft in history.

Match your answer in the solution bank to find the letter for each blank.

Follow the directions for each set:

Simplify:

(394) $(8a^2 - 6a) - (5a^2 - 6a + 2)$

(218) $(7y^3 - y^2 + 4) - (3y^3 + 2y^2 - y)$

(425) $(5x^2 + 3x - 2) - (2x^2 - 4x + 1)$

(155) $(-2z^2 + 5z - 1) - (-z^2 + 3z - 6)$

(473) $(10x^4 - 3x^2) - (4x^4 - 3x^2 + x)$

Translate and simplify:

(629) Subtract $(4w^2 - w)$ from $(6w^2 + 3w - 2)$

(561) Subtract $(2m^2 + m - 3)$ from $(3m^2 - m + 5)$

(776) Subtract $(x^2 - 5x)$ from $(4x^2 + 2x)$

(802) Subtract $(-3p^3 + 2p)$ from $(p^3 - p^2 + 4p)$

Solution Bank

Solution	*Letter*
$m^2 + 5m - 8$	[K]
$6x^4 - x$	[S]
$3a^2 - 2$	[N]
$-z^2 + 2z + 5$	[M]
$3x^2 + 7x - 3$	[G]
$6x^4 + x$	[P]
$-3x^2 - 7x$	[J]
$2w^2 + 4w - 2$	[U]
$4p^3 - p^2 + 2p$	[E]
$3x^2 - x - 1$	[Q]
$m^2 - 2m + 8$	[D]
$4y^3 - 3y^2 + y + 4$	[R]
$3x^2 + 7x$	[A]

___ ___ ___ ___ ___ ___ ___   ___ ___ ___ ___ ___ ___

425 776 218 561 394 802 218   155 629 473 802 629 155

yoursimplemath.com

Name: ____________________ Date: ____________

Add and Subtract Polynomials

In 2004, armed men in ski masks stormed an Oslo museum in broad daylight to steal this iconic, deeply psychological painting by Edvard Munch.

Match your answer in the solution bank to find the letter for each blank.

Follow the directions for each set:

Simplify:

(124) $(-2z^3 + 5z) - (z^3 - z^2 + 3) + (4z^2 - z)$

(910) $(4x^3 - 2x) - (x^2 + 5x) + (3x^3 - 7)$

(605) $(5m^4 - m^2) + (2m^3 - 3m^2 + 4) - (m^4 + m^3 - 2m)$

(781) $(6p^2 - p + 2) + (p^3 - 4p^2) - (2p^3 + 3p - 1)$

Translate and simplify:

(348) Subtract $(2y^2 - 3y + 1)$ from the sum of $(y^3 + 4y^2)$ and $(5y - 2)$

(892) Subtract $(a^3 - 2a^2 + 4)$ from the sum of $(3a^3 - a)$ and $(a^2 + 5a)$

(439) Subtract $(x^4 + 2x^2 - x)$ from the sum of $(4x^4 - 3x^3)$ and $(x^2 + 6)$

(256) Subtract $(5w^2 - w + 3)$ from the sum of $(2w^3 - w^2)$ and $(4w^3 + 2w)$

Solution Bank

Solution	*Letter*
$-3z^3 + 5z^2 + 4z - 3$	[C]
$7x^3 - x^2 + 3x - 7$	[B]
$3x^4 - 3x^3 + 3x^2 - x + 6$	[D]
$6w^3 - 6w^2 + 3w - 3$	[M]
$7x^3 - x^2 - 7x - 7$	[T]
$3x^4 - 3x^3 - x^2 + x + 6$	[R]
$2a^3 + 3a^2 + 4a - 4$	[S]
$y^3 + 2y^2 + 8y - 3$	[H]
$-p^3 + 2p^2 - 4p + 3$	[A]
$2a^3 - a^2 + 4a + 4$	[Y]
$4m^4 + m^3 - 4m^2 + 2m + 4$	[E]
$y^3 + 2y^2 + 2y - 1$	[F]

___ ___ ___   ___ ___ ___ ___ ___ ___

910 348 605   892 124 439 605 781 256

Name: ______________________ Date: ______________

Algebra 1: Beginner

Add and Subtract Polynomials

In 1911, an Italian handyman hid in a broom closet overnight and walked out of the Louvre with this legendary portrait hidden under his smock.

MONA LISA

Follow the directions for each set:

Simplify:

(688) $(-5x^3 + 4x^2) + (3x^3 - 2x^2 - x)$

$-2x^3 + 2x^2 - x$ **688 [L]**

(812) $(3x^2 - 4x) + (2x^2 + 5x)$

$5x^2 + x$ **812 [M]**

(509) $(a^2 - 3a + 2) + (5a^2 + a - 8)$

$6a^2 - 2a - 6$ **509 [N]**

(904) $(7z - 9) + (2z^2 - 3z + 4)$

$2z^2 + 4z - 5$ **904 [S]**

Translate and simplify:

(147) Find the sum of $(2m^2 - m + 4)$ and $(m^2 + 6)$

$3m^2 - m + 10$ **147 [A]**

(234) Find the sum of $(4y^3 + 2y)$ and $(y^3 - 7y + 1)$

$5y^3 - 5y + 1$ **234 [O]**

(351) Find the sum of $(p^4 - 2p^2)$ and $(3p^4 + p^2 + p)$

$4p^4 - p^2 + p$ **351 [I]**

Solution Bank

Solution	*Letter*
$5x^2 + x$	[M]
$6a^2 - 2a - 6$	[N]
$2z^2 + 4z - 5$	[S]
$5y^3 - 5y + 1$	[O]
$6a^2 - 4a - 6$	[Z]
$-2x^3 + 2x^2 - x$	[L]
$4p^4 - p^2 + p$	[I]
$5y^3 + 9y + 1$	[W]
$3m^2 - m + 10$	[A]
$5x^2 - 9x$	[X]

___ ___ ___ ___   ___ ___ ___ ___

812 234 509 147   688 351 904 147

Name: ______________________ Date: ____________

Algebra 1: Intermediate

Add and Subtract Polynomials

In 1990, two thieves dressed as police officers tied up guards and stole $500 million worth of art from this Boston institution, executing the largest property theft in history.

GARDNER MUSEUM

Follow the directions for each set:

Simplify:

(394) $(8a^2 - 6a) - (5a^2 - 6a + 2)$
$3a^2 - 2$ **394 [N]**

(218) $(7y^3 - y^2 + 4) - (3y^3 + 2y^2 - y)$
$4y^3 - 3y^2 + y + 4$ **218 [R]**

(425) $(5x^2 + 3x - 2) - (2x^2 - 4x + 1)$
$3x^2 + 7x - 3$ **425 [G]**

(155) $(-2z^2 + 5z - 1) - (-z^2 + 3z - 6)$
$-z^2 + 2z + 5$ **155 [M]**

(473) $(10x^4 - 3x^2) - (4x^4 - 3x^2 + x)$
$6x^4 - x$ **473 [S]**

Translate and simplify:

(629) Subtract $(4w^2 - w)$ from $(6w^2 + 3w - 2)$
$2w^2 + 4w - 2$ **629 [U]**

(561) Subtract $(2m^2 + m - 3)$ from $(3m^2 - m + 5)$
$m^2 - 2m + 8$ **561 [D]**

(776) Subtract $(x^2 - 5x)$ from $(4x^2 + 2x)$
$3x^2 + 7x$ **776 [A]**

(802) Subtract $(-3p^3 + 2p)$ from $(p^3 - p^2 + 4p)$
$4p^3 - p^2 + 2p$ **802 [E]**

Solution Bank

Solution	*Letter*
$m^2 + 5m - 8$	[K]
$6x^4 - x$	[S]
$3a^2 - 2$	[N]
$-z^2 + 2z + 5$	[M]
$3x^2 + 7x - 3$	[G]
$6x^4 + x$	[P]
$-3x^2 - 7x$	[J]
$2w^2 + 4w - 2$	[U]
$4p^3 - p^2 + 2p$	[E]
$3x^2 - x - 1$	[Q]
$m^2 - 2m + 8$	[D]
$4y^3 - 3y^2 + y + 4$	[R]
$3x^2 + 7x$	[A]

___	___	___	___	___	___	___	___	___	___	___	___	___
425	776	218	561	394	802	218	155	629	473	802	629	155

Name: ______________________ Date: ____________

Algebra 1: Challenge

Add and Subtract Polynomials

In 2004, armed men in ski masks stormed an Oslo museum in broad daylight to steal this iconic, deeply psychological painting by Edvard Munch.

THE SCREAM

Follow the directions for each set:

Simplify:

(124) $(-2z^3 + 5z) - (z^3 - z^2 + 3) + (4z^2 - z)$

$-3z^3 + 5z^2 + 4z - 3$ **124 [C]**

(910) $(4x^3 - 2x) - (x^2 + 5x) + (3x^3 - 7)$

$7x^3 - x^2 - 7x - 7$ **910 [T]**

(605) $(5m^4 - m^2) + (2m^3 - 3m^2 + 4) - (m^4 + m^3 - 2m)$

$4m^4 + m^3 - 4m^2 + 2m + 4$ **605 [E]**

(781) $(6p^2 - p + 2) + (p^3 - 4p^2) - (2p^3 + 3p - 1)$

$-p^3 + 2p^2 - 4p + 3$ **781 [A]**

Translate and simplify:

(348) Subtract $(2y^2 - 3y + 1)$ from the sum of $(y^3 + 4y^2)$ and $(5y - 2)$

$y^3 + 2y^2 + 8y - 3$ **348 [H]**

(892) Subtract $(a^3 - 2a^2 + 4)$ from the sum of $(3a^3 - a)$ and $(a^2 + 5a)$

$2a^3 + 3a^2 + 4a - 4$ **892 [S]**

(439) Subtract $(x^4 + 2x^2 - x)$ from the sum of $(4x^4 - 3x^3)$ and $(x^2 + 6)$

$3x^4 - 3x^3 - x^2 + x + 6$ **439 [R]**

(256) Subtract $(5w^2 - w + 3)$ from the sum of $(2w^3 - w^2)$ and $(4w^3 + 2w)$

$6w^3 - 6w^2 + 3w - 3$ **256 [M]**

Solution Bank

Solution	*Letter*
$-3z^3 + 5z^2 + 4z - 3$	[C]
$7x^3 - x^2 + 3x - 7$	[B]
$3x^4 - 3x^3 + 3x^2 - x + 6$	[D]
$6w^3 - 6w^2 + 3w - 3$	[M]
$7x^3 - x^2 - 7x - 7$	[T]
$3x^4 - 3x^3 - x^2 + x + 6$	[R]
$2a^3 + 3a^2 + 4a - 4$	[S]
$y^3 + 2y^2 + 8y - 3$	[H]
$-p^3 + 2p^2 - 4p + 3$	[A]
$2a^3 - a^2 + 4a + 4$	[Y]
$4m^4 + m^3 - 4m^2 + 2m + 4$	[E]
$y^3 + 2y^2 + 2y - 1$	[F]

___ ___ ___ ___ ___ ___ ___ ___ ___
910 348 605 892 124 439 605 781 256